Z80 mnemonics and hex codes

The mnemonics and hex codes for the instructions covered in this book are given on the next few pages. The term "implicit addressing" used in these lists is just the name for instructions where no operand need be specified in the hex code. There are a few other instructions not listed here and if you want to go further with machine code you will need a complete list of the Z80 instruction set (see page 40). The following abbreviations are used in these lists:

n = number
nn = two byte number
r = register
rr = register pair
x = address
c = condition
d = displacement

ADC A,n Add with carry, a number, n, to the accumulator. (Immediate addressing.)

ADC A,n	CE,n

ADC A,r Add with carry, register r to the accumulator. (Implicit addressing.)

ADC A,A	8F
ADC A,B	88
ADC A,C	89
ADC A,D	8A
ADC A,E	8B
ADC A,H	8C
ADC A,L	8D

ADC HL,rr Add with carry, the contents of register pair rr to HL. (Implicit addressing.)

ADC HL,BC	ED4A
ADC HL,DE	ED5A
ADC HL,HL	ED6A

ADD A,n Add a number, n, to the accumulator. (Immediate addressing.)

ADD,n	C6,n

ADD A,r Add register r to the accumulator. (Implicit addressing.)

ADD A,A	87
ADD A,B	80
ADD A,C	81
ADD A,D	82
ADD A,E	83
ADD A,H	84
ADD A,L	85

ADD HL,rr Add the contents of register pair rr to HL. (Implicit addressing.)

ADD HL,BC	09
ADD HL,DE	19
ADD HL,HL	29

CALL x Go to subroutine starting at address x. (Immediate addressing.)

CALL x	CD x

CALL c,x Go to subroutine starting at address x depending on condition c. c may be Z (equal); NZ (not equal); C (carry); NC (no carry); PE (parity even); PO (parity odd); M (minus) or P (plus). (Immediate addressing.)

CALL Z,x	CC,x
CALL NZ,x	C4,x
CALL C,x	DC,x
CALL NC,x	D4,x
CALL PE,x	EC,x
CALL PO,x	E4,x
CALL M,x	FC,x
CALL P,x	F4,x

CCF Complement carry flag. (Implicit addressing.)

CCF	3F

CP n Compare contents of accumulator with data n. (Immediate addressing.)

CP n	FE n

CP r Compare contents of register r with the accumulator. (Implicit addressing.)

CP A	BF
CP B	B8
CP C	B9
CP D	BA
CP E	BB
CP H	BC
CP L	BD

CP (HL) Compare contents of accumulator with contents of address held in HL. (Indirect addressing.)

CP (HL)	BE

DEC r Decrement register r. (Implicit addressing.)

DEC A	3D
DEC B	05
DEC C	0D
DEC D	15
DEC E	1D
DEC H	25
DEC L	2D

DEC rr Decrement register pair rr. (Implicit addressing.)

DEC BC	0B
DEC DE	1B
DEC HL	2B
DEC IX	DD2B
DEC IY	FD2B

DEC (HL) Decrement contents of address held in HL. (Indirect addressing.)

DEC (HL)	35

INC r Increment register r. (Implicit addressing.)

INC A	3C
INC B	04
INC C	0C
INC D	14
INC E	1C
INC H	24
INC L	2C

INC rr Increment register pair rr. (Implicit addressing.)

INC BC	03
INC DE	13
INC HL	23

INC (HL) Increment contents of address held in HL. (Indirect addressing.)

INC (HL)	34

JP x Jump to address x. (Immediate addressing.)

JP x	C3 x

JP (rr) Jump to address held in register pair rr. (Implicit addressing.)

JP (HL)	E9
JP (IX)	DDE9
JP (IY)	FDE9

JP c,x Jump to address x depending on condition c. c may be Z (equal); NZ (not equal); C (carry); NC (no carry); PE (parity even); PO (parity odd); M (minus) or P (plus). (Immediate addressing.)

JP Z,x	CA,x
JP NZ,x	C2,x
JP C,x	DA,x
JP NC.x	D2,x
JP PE,x	EA,x
JP PO,x	E2,x
JP M,x	FA,x
JP P,x	F2,x

JR d Jump relative. Jump d bytes (the displacement). (Relative addressing.)

JR d	18 d

JR c,d Jump relative. Jump d bytes (the displacement) depending on condition c. c may be NZ (not equal); Z (equal); NC (no carry) or C (carry). (Relative addressing.)

JR NZ,d	20,d
JR Z,d	28,d
JR NC,d	30,d
JR C,d	38,d

LD r,n Load register r with data n. (Immediate addressing.)

LD A,n	3E,n
LD B,n	06,n
LD C,n	0E,n
LD D,n	16,n
LD E,n	1E,n
LD H,n	26,n
LD L,n	2E,n

LD rr,nn Load register pair rr with two byte number nn. (Immediate addressing.)

LD BC,nn	01,nn
LD DE,nn	11,nn
LD HL,nn	21,nn

LD A, (x) Load accumulator with contents of address x. (Absolute addressing.)

LD A, (x)	3A, (x)

LD rr, (x) Load register pair rr with contents of addresses x and x+1. (Absolute addressing.)

LD BC, (x)	ED4B, (x)
LD DE, (x)	ED5B, (x)
LD HL, (x)	2A, (x)

LD A,r Load the accumulator with contents of register r. (Implicit addressing.)

LD A,A	7F
LD A,B	78
LD A,C	79
LD A,D	7A
LD A,E	7B
LD A,H	7C
LD A,L	7D

LD B,r Load register B with the contents of register r. (Implicit addressing.)

LD B,A	47
LD B,B	40
LD B,C	41
LD B,D	42
LD B,E	43
LD B,H	44
LD B,L	45

LD C,r Load register C with the contents of register r. (Implicit addressing.)

LD C,A	4F
LD C,B	48
LD C,C	49
LD C,D	4A
LD C,E	4B
LD C,H	4C
LD C,L	4D

LD D,r Load register D with the contents of register r. (Implicit addressing.)

LD D,A	57
LD D,B	50
LD D,C	51
LD D,D	52
LD D,E	53
LD D,H	54
LD D,L	55

LD E,r Load register E with the contents of register r. (Implicit addressing.)

LD E,A	5F
LD E,B	58
LD E,C	59
LD E,D	5A
LD E,E	5B
LD E,H	5C
LD E,L	5D

LD H,r Load register H with the contents of register r. (Implicit addressing.)

LD H,A	67
LD H,B	60
LD H,C	61
LD H,D	62
LD H,E	63
LD H,H	64
LD H,L	65

LD L,r Load register L with the contents of register r. (Implicit addressing.)

LD L,A	6F
LD L,B	68
LD L,C	69
LD L,D	6A
LD L,E	6B
LD L,H	6C
LD L,L	6D

LD r, (rr) Load register r with contents of address held in register pair rr. (Indirect addressing.)

LD A, (BC)	0A
LD A,(DE)	1A
LD A,(HL)	7E
LD B,(HL)	46
LD C,(HL)	4E
LD D,(HL)	56
LD E,(HL)	5E
LD H,(HL)	66
LD L,(HL)	6E

LD (x),A Store the contents of the accumulator in address x. (Absolute addressing.)

LD (x),A	32,x

LD (x),rr Store the contents of register pair rr at addresses x and x+1. (Absolute addressing.)

LD (x),BC	ED43,x
LD (x),DE	ED53,x
LD (x),HL	22,x

LD (rr),r Store the contents of register r at the address held in register pair rr. (Indirect addressing.)

LD (BC),A	02
LD (DE),A	12
LD (HL),A	77
LD (HL),B	70
LD (HL),C	71
LD (HL),D	72
LD (HL),E	73
LD (HL),H	74
LD (HL),L	75

LD (rr), n Store data n at address held in register pair rr. (Immediate/indirect addressing.)

LD (HL),n	36

RET Return from subroutine. (Indirect addressing.)

RET	C9

RET c Return from subroutine depending on condition c. c can be Z (equal); NZ (not equal); C (carry); NC (no carry); PE (parity even); PO (parity odd); P (plus); M (minus). (Indirect addressing.)

RET Z	C8
RET NZ	C0
RET C	D8
RET NC	D0
RET PE	E8
RET PO	E0
RET M	F8
RET P	F0

SBC A,n Subtract with carry data n from the accumulator. (Immediate addressing.)

SBC A,n	DE,n

SBC A,r Subtract with carry contents of register r from the accumulator. (Implicit addressing.)

SBC A,A	9F
SBC A,B	98
SBC A,C	99
SBC A,D	9A
SBC A,E	9B
SBC A,H	9C
SBC A,L	9D

SBC HL,rr Subtract with carry contents of register pair rr from register pair HL. (Implicit addressing.)

SBC HL,BC	ED42
SBC HL,DE	ED52
SBC HL,HL	ED62

SBC A,(HL) Subtract with carry the contents of address held in register pair HL, from the accumulator. (Indirect addressing).

SBC A,(HL)	9E

SCF Set carry flag. (Implicit addressing.)

SCF	37

SUB n Subtract data n from the accumulator. (Immediate addressing.)

SUB, n	D6, n

SUB r Subtract contents of register r from the accumulator. (Immediate addressing.)

SUB A	97
SUB B	90
SUB C	91
SUB D	92
SUB E	93
SUB H	94
SUB L	95

SUB (HL) Subtract the contents of address held in HL from the accumulator. (Indirect addressing.)

SUB (HL)	96

Puzzle answers

Page 11

&A7 in decimal is 167. 513 in hex is &201.

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Tip: an easy way to work out the two's complement of a number is to subtract it from 256, then convert the answer to hex. E.g. 256 − 6 = 250 which is FA in hex.

1. 25+73 (25 is &19 and 73 is &49)

Z80		6502		Meaning
Mnemonics	**Hex codes**	**Mnemonics**	**Hex codes**	
LD A, &19	3E,19	LDA #&19	A9 19	Put &19 in accumulator.
ADD A, &49	C6,49	ADC #&49	69 49	Add &49 to accumulator.
LD (address), A	32, address	STA address	8D address	Store contents of accumulator at a certain address.
RET	C9	RTS	60	Return

2. 64+12+14 (64 is &40, 12 is &0C and 14 is &0E)

Z80		6502		Meaning
Mnemonics	**Hex codes**	**Mnemonics**	**Hex codes**	
LD A, &40	3E,40	LDA #&40	A9 40	Put &40 in accumulator.
ADD A, &0C	C6,0C	ADC #&0C	69 0C	Add &0C to accumulator.
ADD A, &0E	C6,0E	ADC #&0E	69 0E	Add &0E to accumulator.
LD(address), A	32, address	STA address	8D address	Store contents of accumulator at a certain address.
RET	C9	RTS	60	Return

Puzzle answers continued on page 48.